

SAFE BAND

AI-POWERED SMART HEALTH MONITORING SYSTEM

R26-IT-057



INTRODUCTION

- **Safe Band** is a multi-component IoT health platform built around an Android smartwatch.
- It continuously monitors patient vitals, detects emergencies using Machine Learning, dispatches ambulances automatically, and supports clinical staff through intelligent dashboards all designed specifically for Sri Lanka's healthcare landscape.



**Emergency
Detection**



**Smart Ambulance
Dispatch**



**Ward Monitoring
& Alerts**



**Heart Risk
Prediction**

PROBLEMS

- Emergency healthcare systems in Sri Lanka still rely on manual processes and delayed communication
- **Most existing healthcare solutions:**
 1. Provide only basic monitoring
 2. Lack real-time emergency response
 3. Are not connected with hospitals and ambulance services
- Patients with critical conditions are not continuously monitored.
- Ambulance dispatch and hospital coordination are often delayed.
- Existing smart wearables mainly focus on fitness tracking, not emergency healthcare.
- Hospitals face difficulties in monitoring multiple patients efficiently.
- Lack of AI-powered predictive healthcare and centralized emergency management systems.



IMPACT / GRAVITY

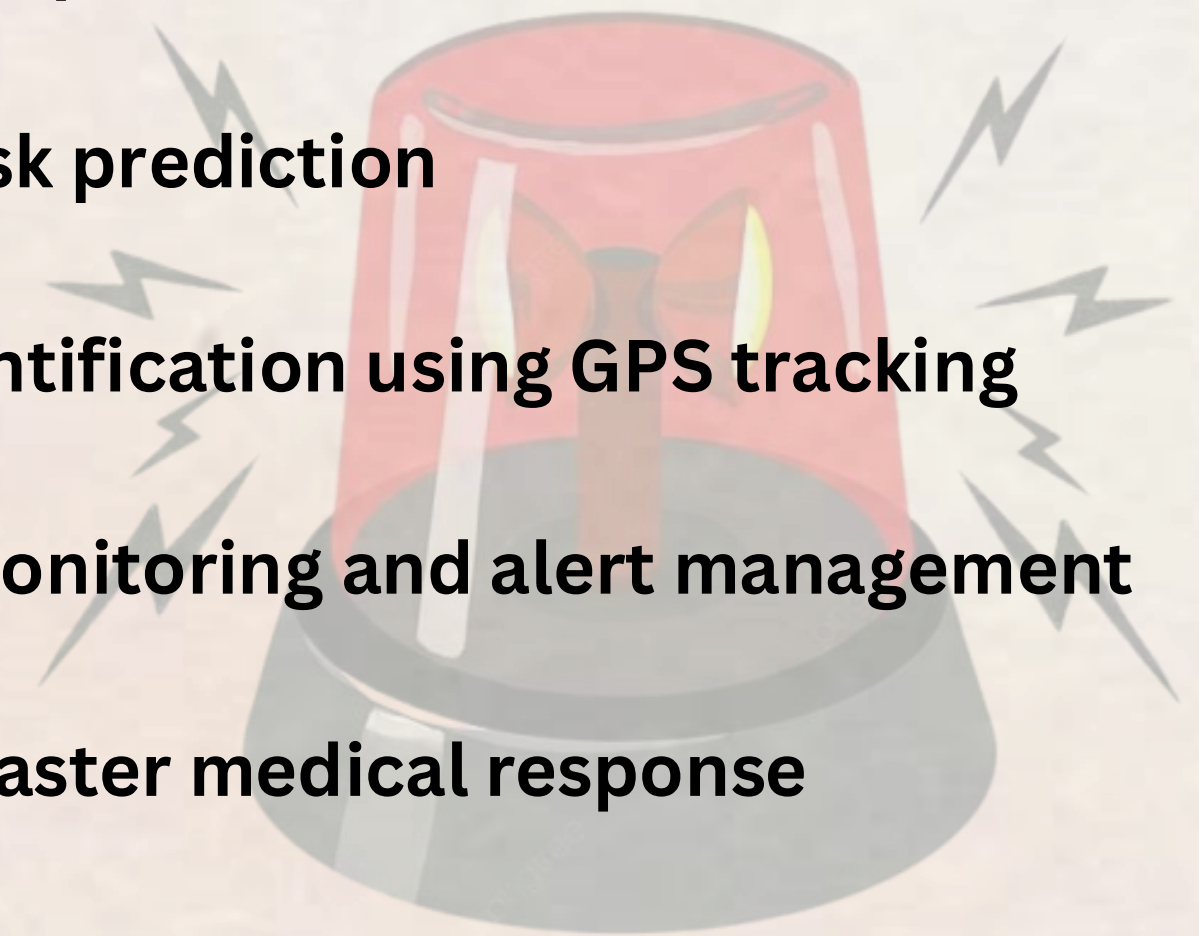
- Delayed emergency response increases the risk of preventable deaths.
- Lack of continuous monitoring reduces early detection of critical conditions.
- Difficulty in tracking patient health conditions for hospitals and healthcare staff.
- Manual ambulance dispatch reduces emergency response efficiency.
- Lack of predictive healthcare systems increases pressure on hospitals.
- AI-powered monitoring improves faster emergency response and patient safety.
- Real-time alerts and hospital notifications enhance healthcare coordination.

TARGET USERS & BENEFICIARIES

- **Primary Users:** Cardiac patients, elderly individuals, and patients with chronic health conditions
- **Secondary Users:** Doctors, nurses, ambulance staff, and emergency healthcare teams
- **Institutions / Society:** Hospitals, emergency response services, and the Sri Lankan healthcare sector
- **Technology Need:** AI-powered IoT healthcare system for real-time monitoring, emergency detection, and smart ambulance dispatch

SOLUTIONS

- Single integrated platform for patient monitoring, emergency detection, hospital notification, and ambulance dispatch
- Real-time monitoring of vital signs including Heart Rate, BP, SpO2, and ECG
- Machine Learning-based emergency detection and health risk prediction
- Smart ambulance dispatch system with nearest hospital identification using GPS tracking
- Centralized hospital ward dashboard for real-time patient monitoring and alert management
- Automated emergency alerts and hospital notifications for faster medical response
- Predictive healthcare analytics to identify critical conditions before emergencies occur
- Cloud-based scalable architecture for future healthcare system expansion and integration



ML DRIVEN EMERGENCY DETECTION SYSTEM



IT22180070 | Dewapriya K.M.S.S | R26-IT-
057

PROBLEMS

NO REAL-TIME DETECTION

- Existing wearables only collect data
- They lack embedded clinical intelligence to classify emergencies autonomously.

SINGLE-PARAMETER LIMITS

- Commercial devices detect one signal (e.g., AF via ECG)
- but cannot identify complex multi-vital emergencies

NO EMERGENCY INTEGRATION

- No current wearable integrates with hospital dashboards or ambulance dispatch.
- causing critical delays in Sri Lanka's healthcare ecosystem.

SOLUTION

MULTI-PARAMETER ML PIPELINE

Fuses HR, BP, SpO₂ & ECG into a unified LSTM/Random Forest classifier detects composite emergencies single-signal devices cannot.

ON-DEVICE REAL-TIME INFERENCE

TensorFlow Lite embedded in the Android Smartwatch app enables near-real-time classification without cloud dependency low latency alerts within seconds.

CLOUD ALERT & DISPATCH INTEGRATION

Firebase Realtime Database propagates critical events instantly to hospital dashboards and the ML Ambulance Dispatch system closing the response gap.

USER REQUIREMENTS

- The system shall continuously monitor HR, BP, SpO₂, and ECG data in real time.
- The system shall detect emergency conditions and generate instant alerts using machine learning.
- The system shall provide a real-time Android dashboard with secure Firebase synchronization.



DESIGN EXCELLENCE & CONTRIBUTION

This ML-driven engine continuously monitors patient vital signs and uses a Random Forest model to detect abnormal health conditions in real time. It automatically classifies patient status as **Normal** or **Emergency** and generates instant alerts for critical situations

NORMAL

```
Columns: Index(['Patient Number', 'Heart Rate (bpm)', 'SpO2 Level (%)',
               'Systolic Blood Pressure (mmHg)', 'Diastolic Blood Pressure (mmHg)',
               'Body Temperature (°C)', 'Fall Detection', 'Predicted Disease',
               'Data Accuracy (%)', 'Heart Rate Alert', 'SpO2 Level Alert',
               'Blood Pressure Alert', 'Temperature Alert'],
              dtype='object')

New Status Column Created!
      Status
0    Normal
1  Emergency
2  Emergency
3  Emergency
4  Emergency

✅ Model Training Completed!

Accuracy: 0.9999

Report:
      precision  recall  f1-score  support
0           1.00     1.00     1.00     9494
1           1.00     1.00     1.00      506

accuracy          1.00     1.00     1.00    10000
macro avg         1.00     1.00     1.00    10000
weighted avg      1.00     1.00     1.00    10000

=====
MODEL PREDICTION: Normal
=====
```

EMERGENCY

```
Columns: Index(['Patient Number', 'Heart Rate (bpm)', 'SpO2 Level (%)',
               'Systolic Blood Pressure (mmHg)', 'Diastolic Blood Pressure (mmHg)',
               'Body Temperature (°C)', 'Fall Detection', 'Predicted Disease',
               'Data Accuracy (%)', 'Heart Rate Alert', 'SpO2 Level Alert',
               'Blood Pressure Alert', 'Temperature Alert'],
              dtype='object')

New Status Column Created!
      Status
0    Normal
1  Emergency
2  Emergency
3  Emergency
4  Emergency

✅ Model Training Completed!

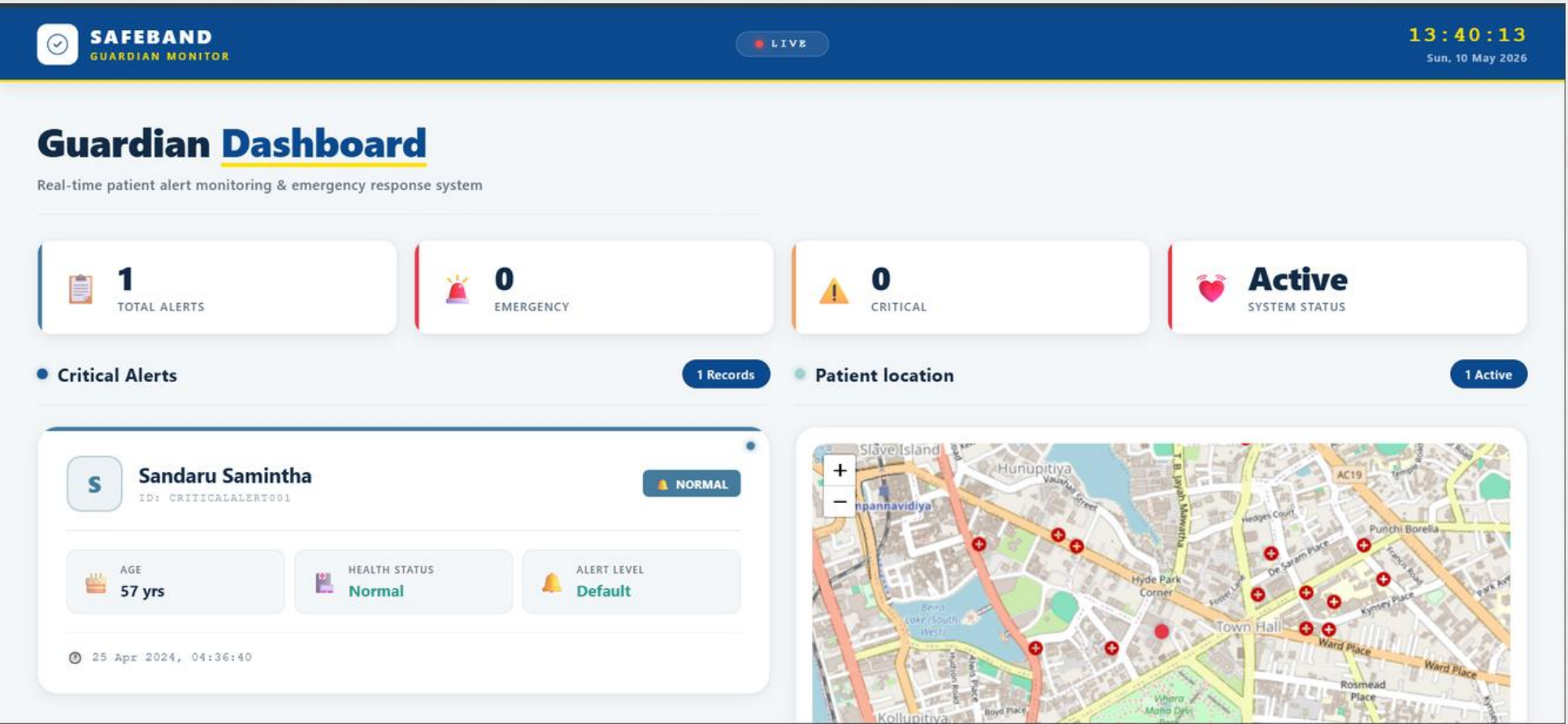
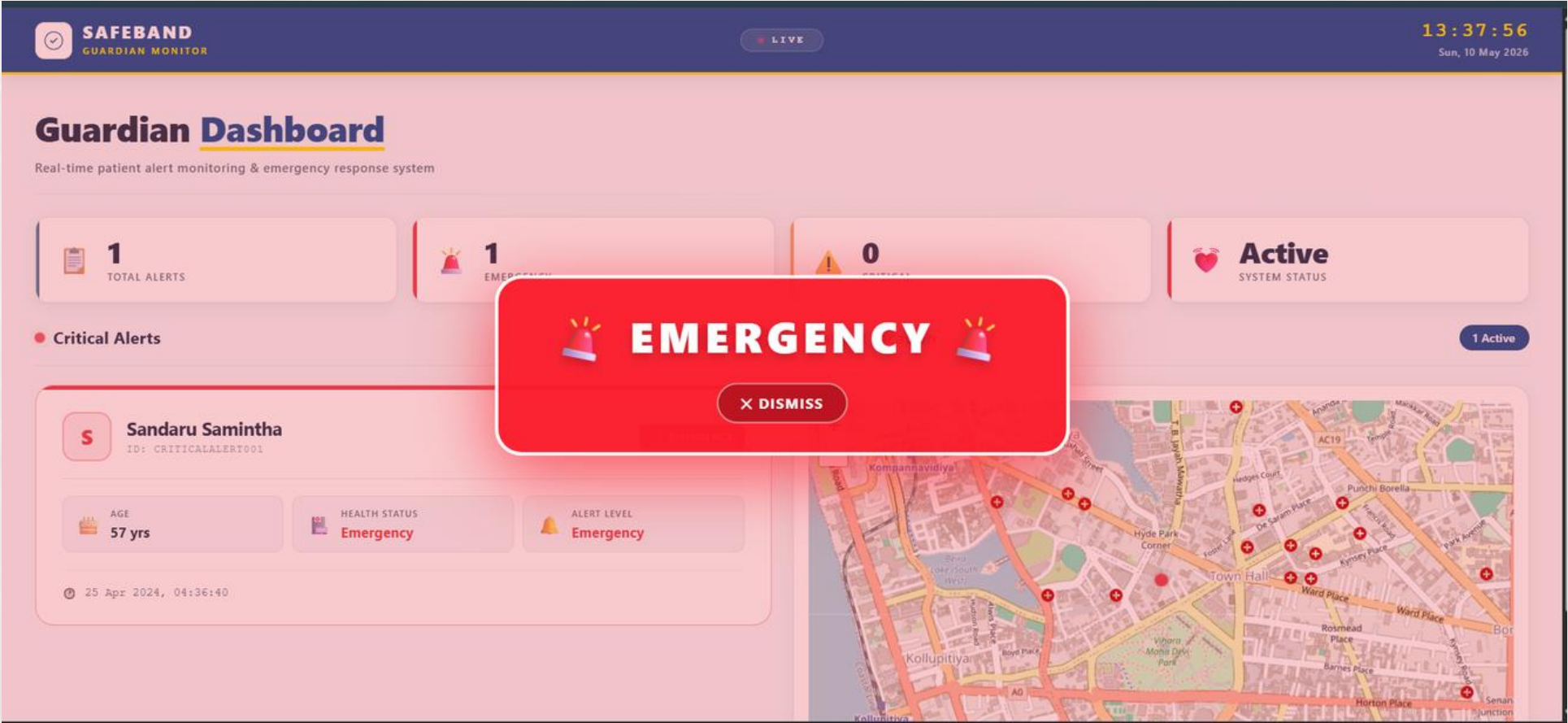
Accuracy: 0.9999

Report:
      precision  recall  f1-score  support
0           1.00     1.00     1.00     9494
1           1.00     1.00     1.00      506

accuracy          1.00     1.00     1.00    10000
macro avg         1.00     1.00     1.00    10000
weighted avg      1.00     1.00     1.00    10000

=====
MODEL PREDICTION: Emergency
=====
```

DESIGN EXCELLENCE & CONTRIBUTION

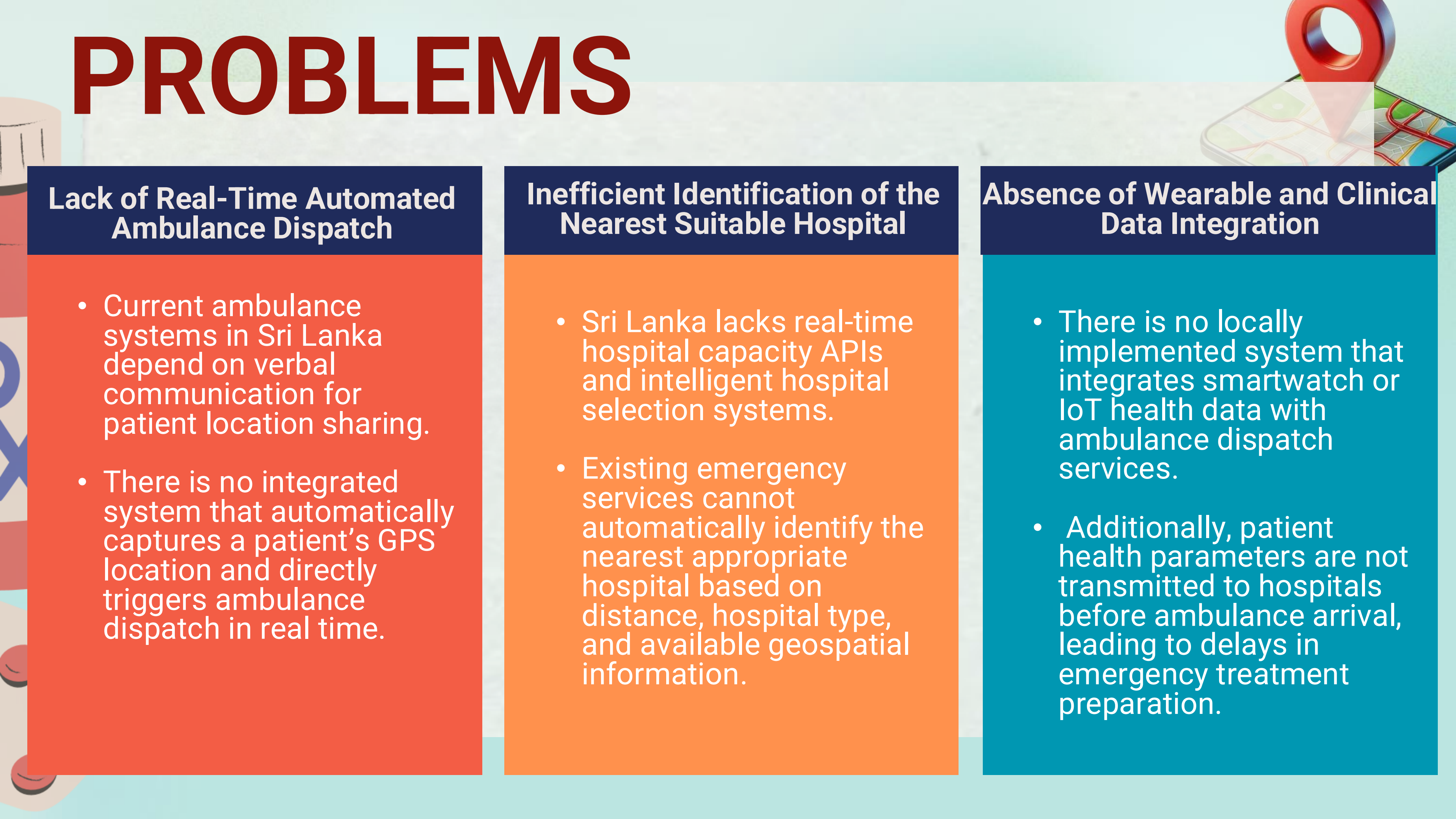


ML-BASED SMART AMBULANCE DISPATCH SYSTEM WITH NEAREST HOSPITAL IDENTIFICATION

IT22101792 | DIMASHA S.S.D. | R26-IT-057



PROBLEMS



Lack of Real-Time Automated Ambulance Dispatch

- Current ambulance systems in Sri Lanka depend on verbal communication for patient location sharing.
- There is no integrated system that automatically captures a patient's GPS location and directly triggers ambulance dispatch in real time.

Inefficient Identification of the Nearest Suitable Hospital

- Sri Lanka lacks real-time hospital capacity APIs and intelligent hospital selection systems.
- Existing emergency services cannot automatically identify the nearest appropriate hospital based on distance, hospital type, and available geospatial information.

Absence of Wearable and Clinical Data Integration

- There is no locally implemented system that integrates smartwatch or IoT health data with ambulance dispatch services.
- Additionally, patient health parameters are not transmitted to hospitals before ambulance arrival, leading to delays in emergency treatment preparation.

SOLUTION

REAL-TIME GPS CAPTURE

- Android smartwatch acquires live patient coordinates on emergency trigger.
- no manual location reporting needed.

AUTOMATED DISPATCH

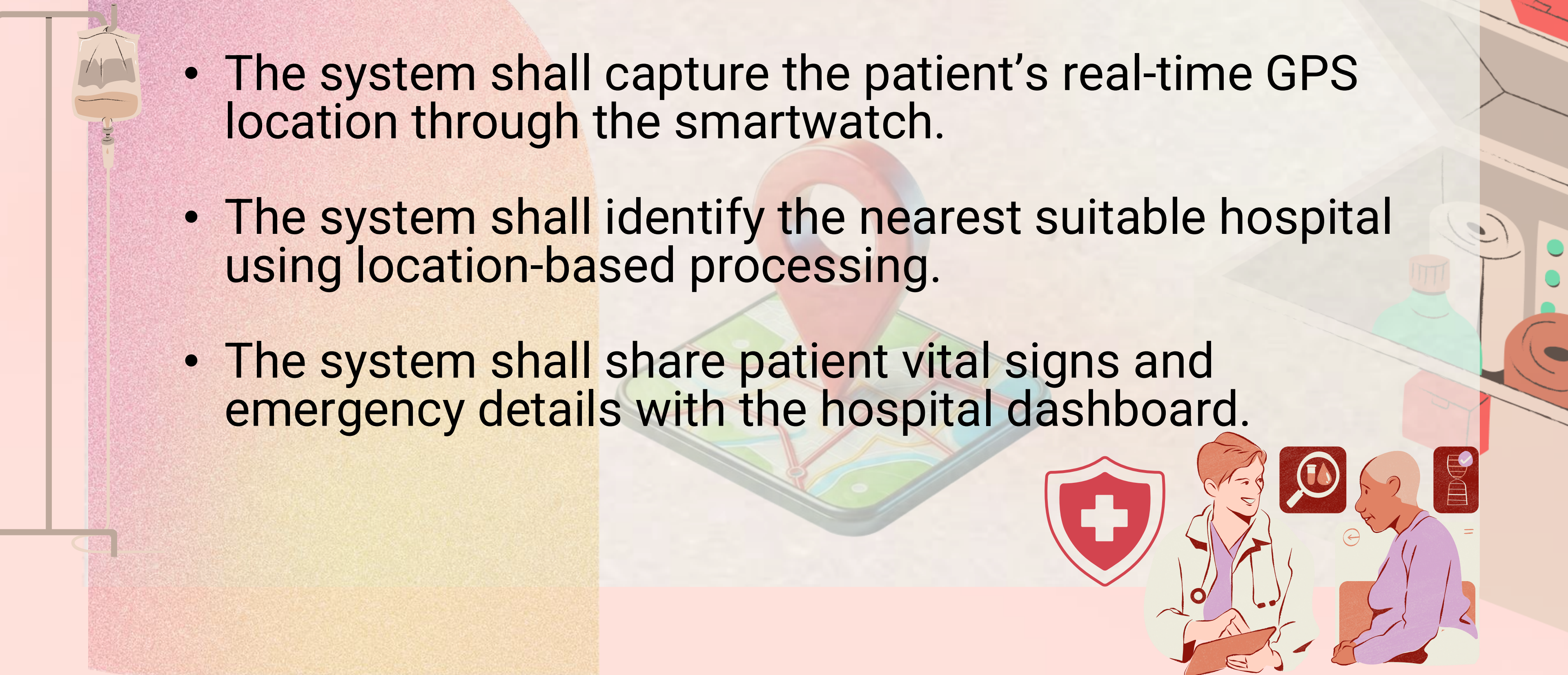
- Ambulance dispatch is triggered automatically with full situational data reducing response latency across Sri Lanka.

HOSPITAL SELECTION

- ML algorithm identifies the nearest suitable hospital using geospatial & registry data instantly.

USER REQUIREMENTS

- The system shall capture the patient's real-time GPS location through the smartwatch.
- The system shall identify the nearest suitable hospital using location-based processing.
- The system shall share patient vital signs and emergency details with the hospital dashboard.



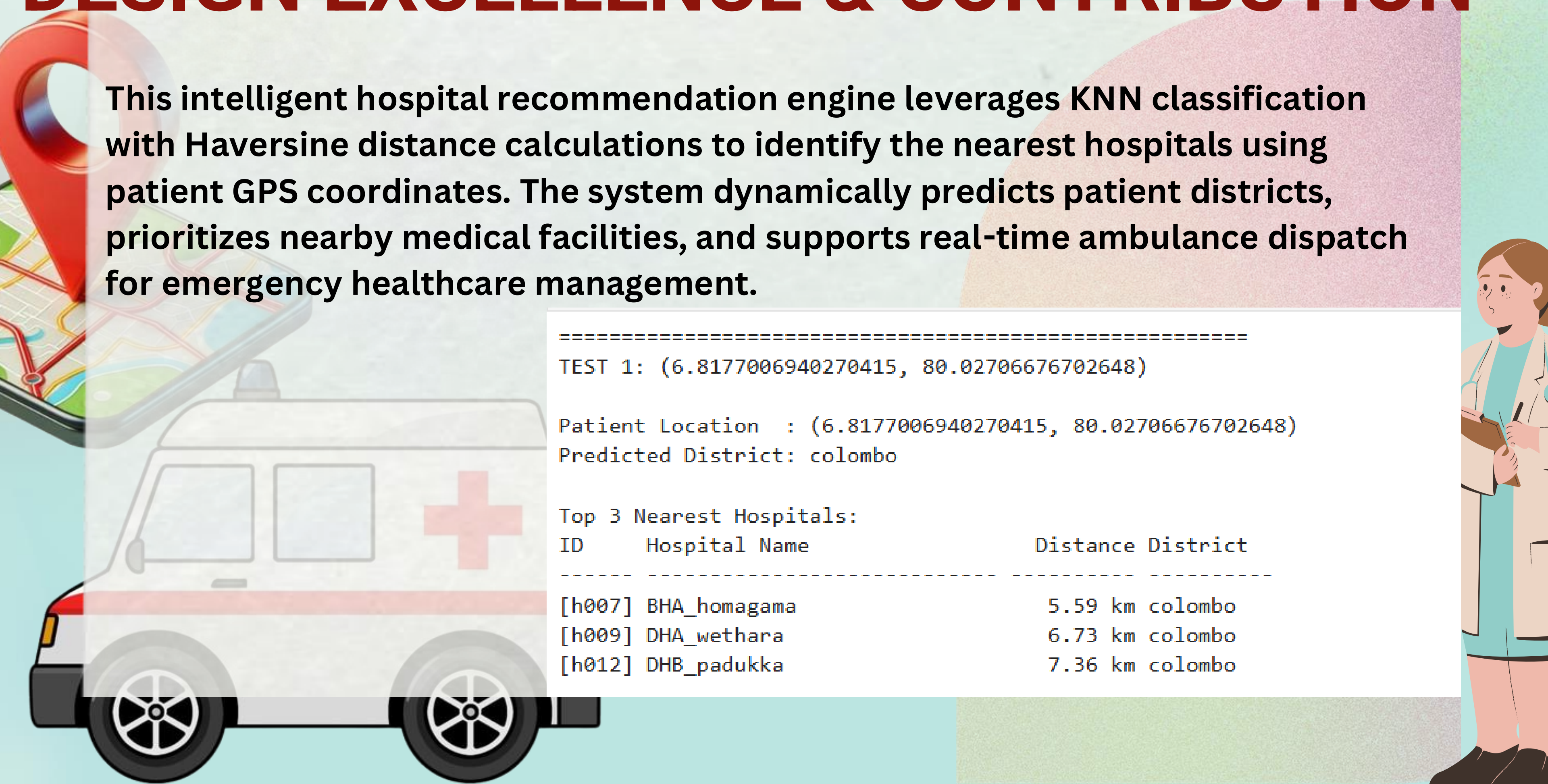
DESIGN EXCELLENCE & CONTRIBUTION

This intelligent hospital recommendation engine leverages KNN classification with Haversine distance calculations to identify the nearest hospitals using patient GPS coordinates. The system dynamically predicts patient districts, prioritizes nearby medical facilities, and supports real-time ambulance dispatch for emergency healthcare management.

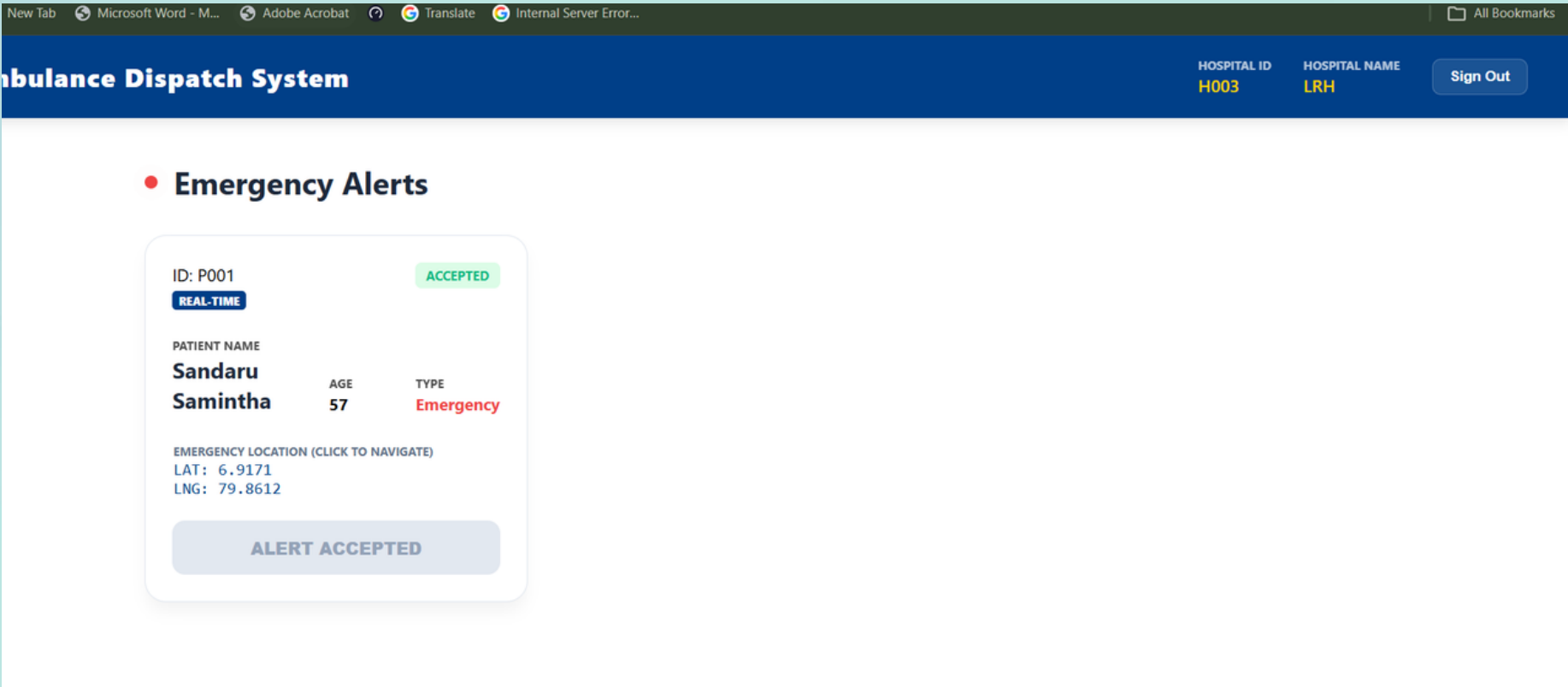
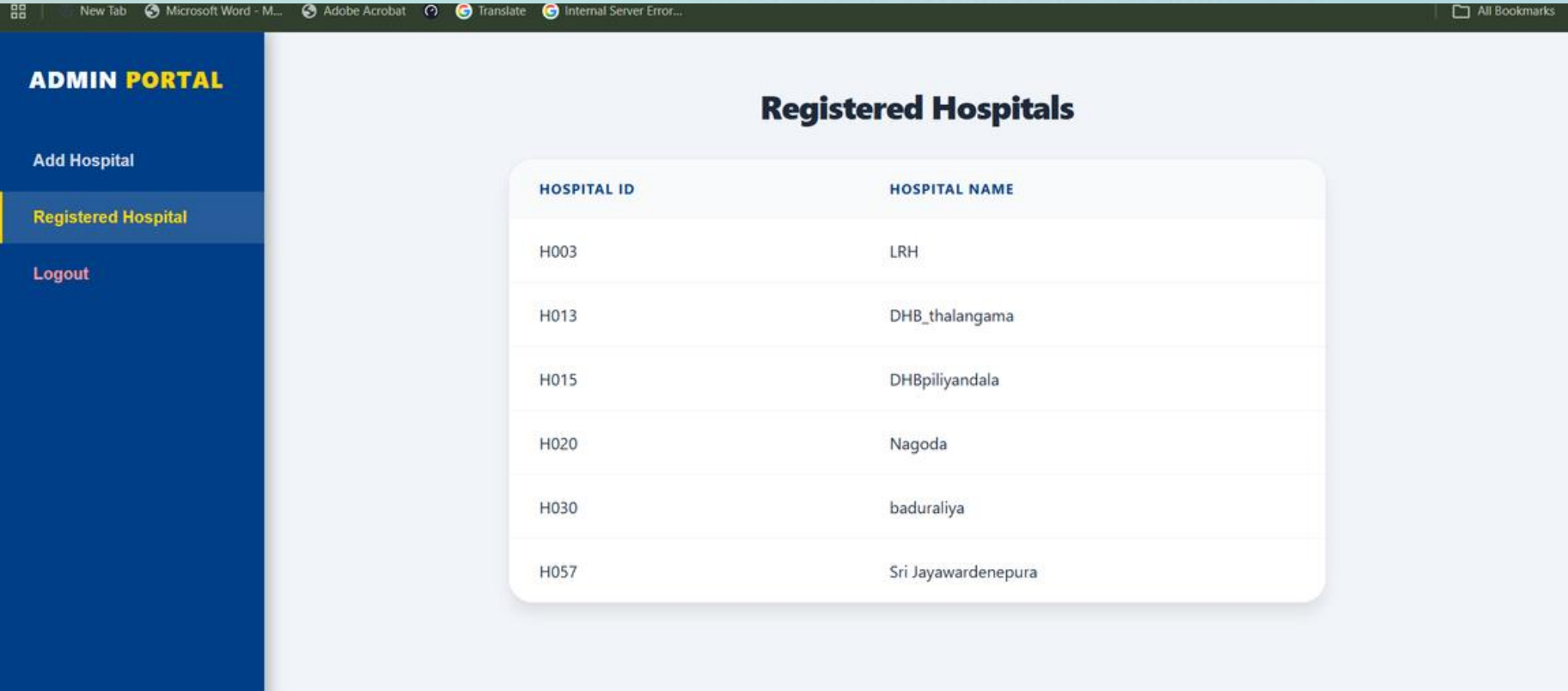
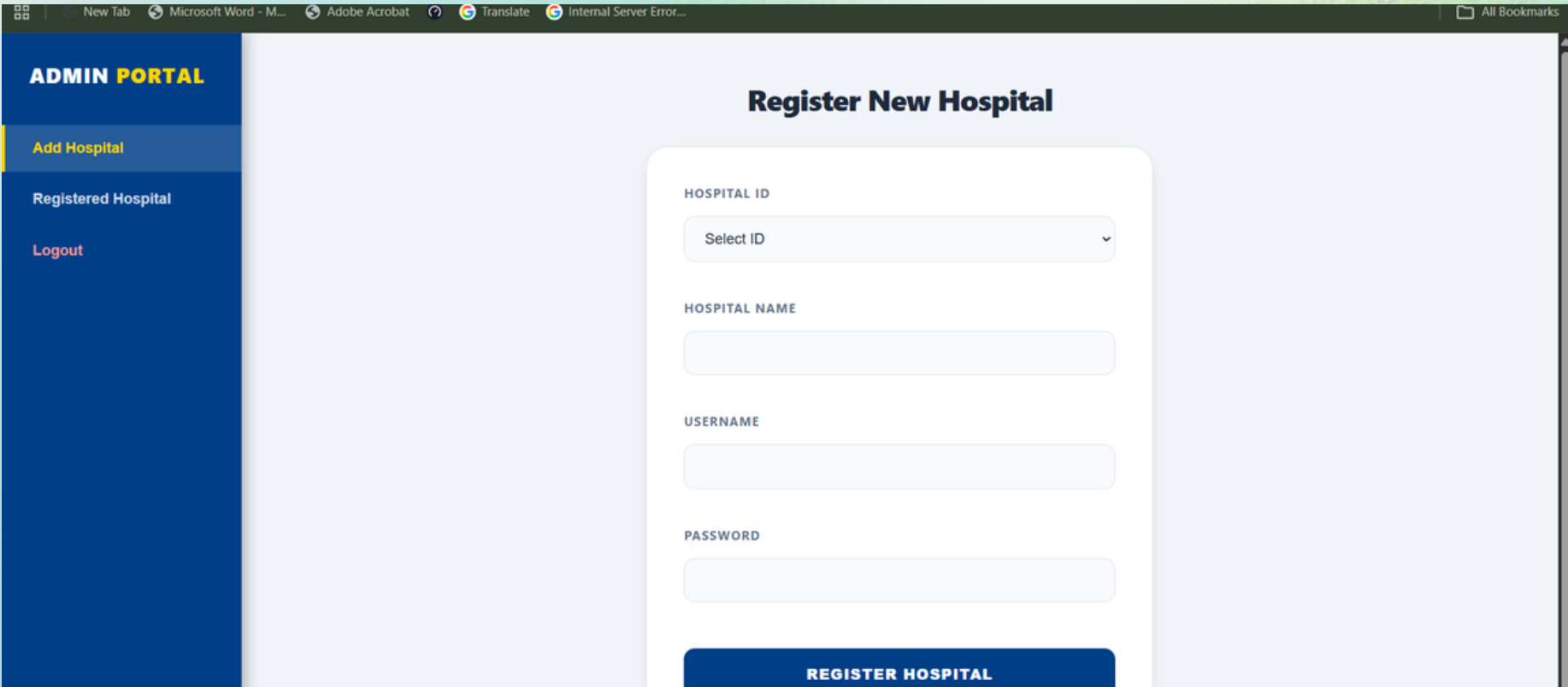
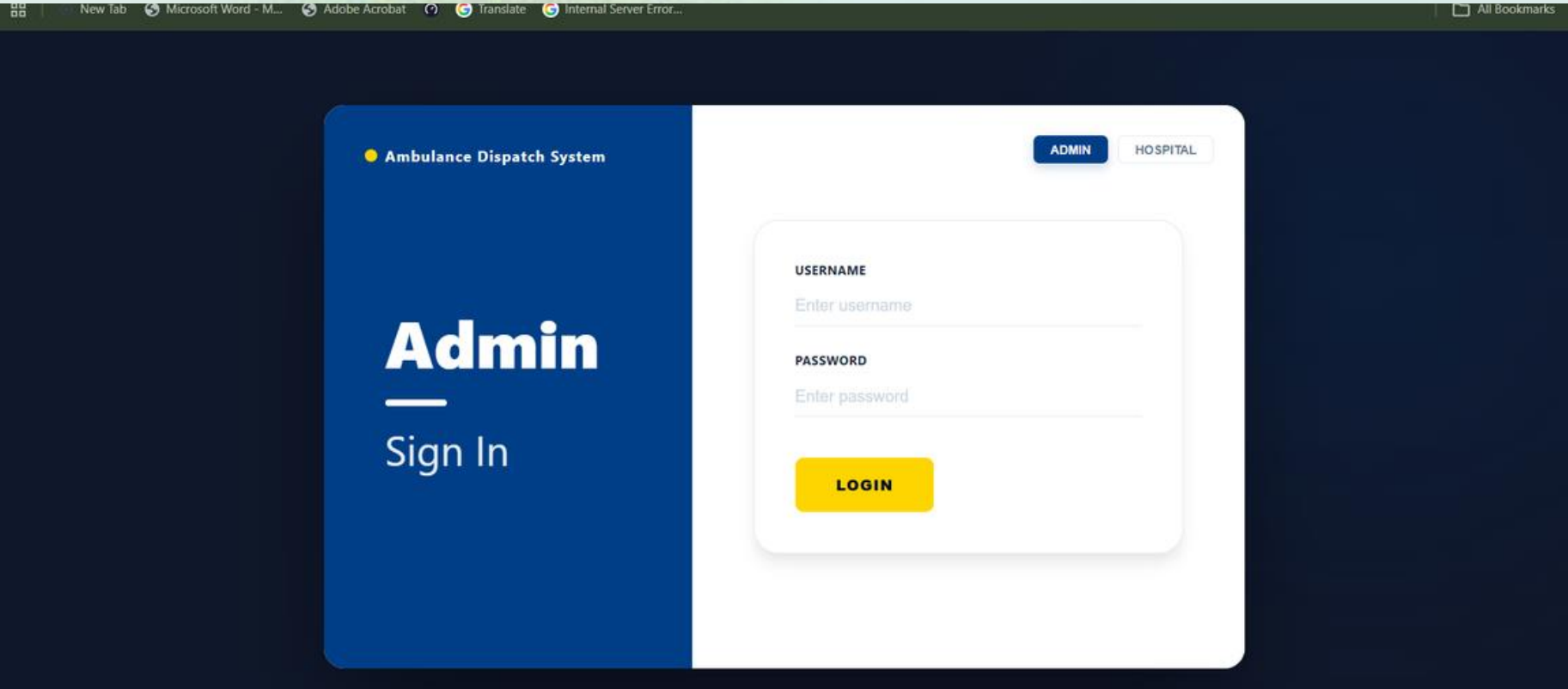
```
=====
TEST 1: (6.8177006940270415, 80.02706676702648)

Patient Location : (6.8177006940270415, 80.02706676702648)
Predicted District: colombo

Top 3 Nearest Hospitals:
ID      Hospital Name      Distance District
-----
[h007] BHA_homagama      5.59 km colombo
[h009] DHA_wethara      6.73 km colombo
[h012] DHB_padukka      7.36 km colombo
```

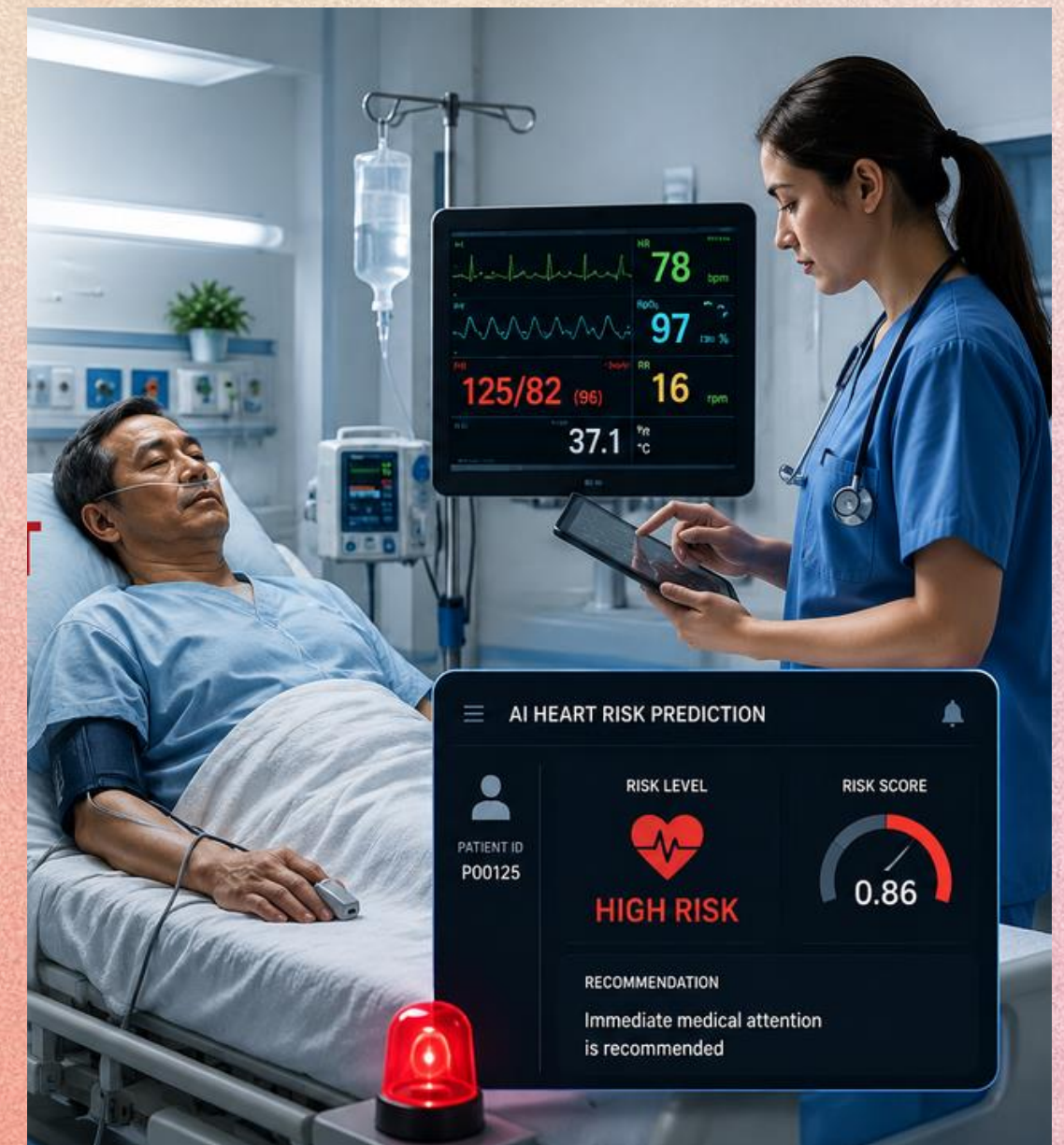


DESIGN EXCELLENCE & CONTRIBUTION

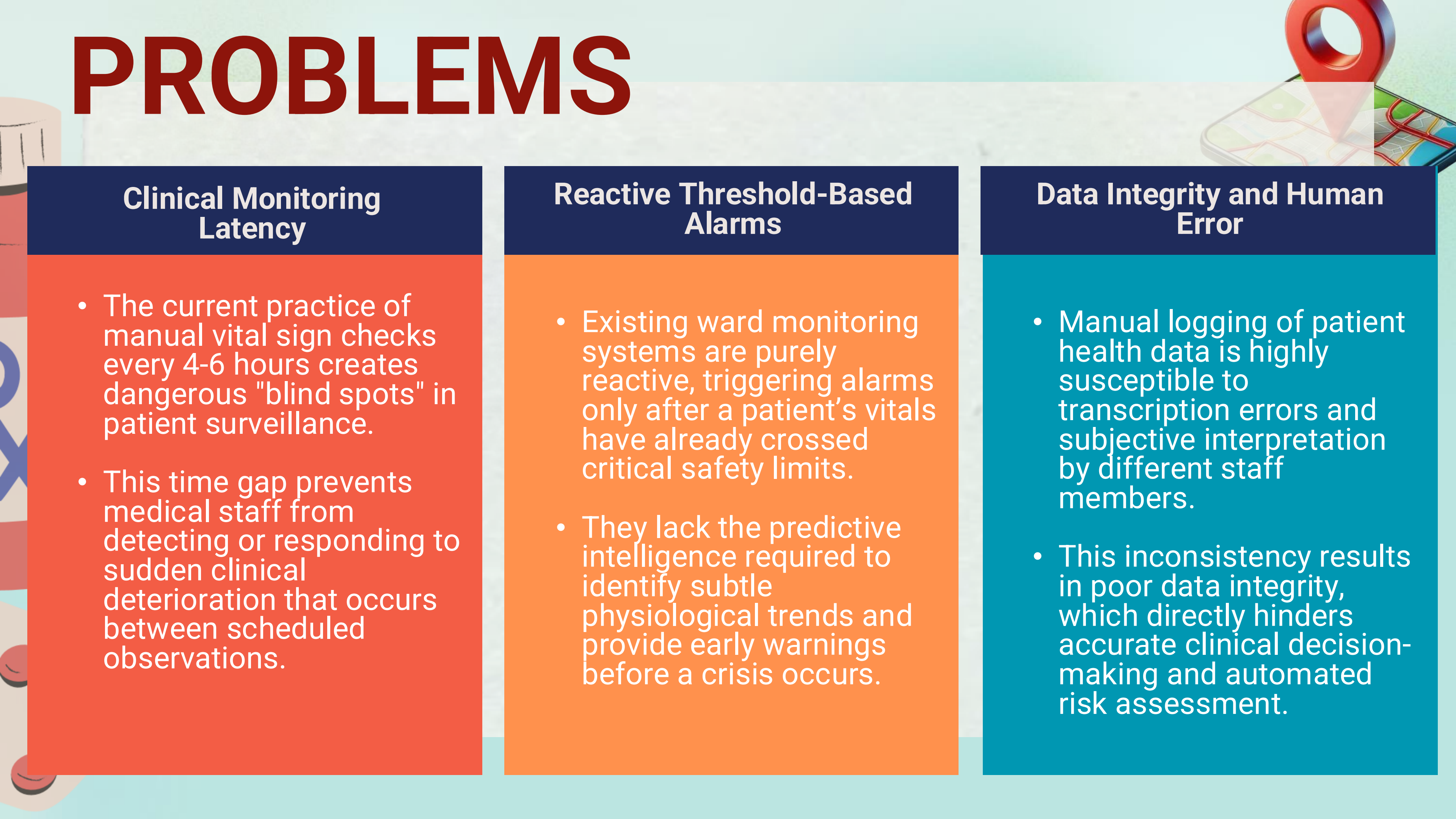


INTELLIGENT AI-BASED HOSPITAL WARD MONITORING AND ALERT MANAGEMENT SYSTEM

IT22190352 | Gamage S.Y.S | R26-IT-
057



PROBLEMS



Clinical Monitoring Latency

- The current practice of manual vital sign checks every 4-6 hours creates dangerous "blind spots" in patient surveillance.
- This time gap prevents medical staff from detecting or responding to sudden clinical deterioration that occurs between scheduled observations.

Reactive Threshold-Based Alarms

- Existing ward monitoring systems are purely reactive, triggering alarms only after a patient's vitals have already crossed critical safety limits.
- They lack the predictive intelligence required to identify subtle physiological trends and provide early warnings before a crisis occurs.

Data Integrity and Human Error

- Manual logging of patient health data is highly susceptible to transcription errors and subjective interpretation by different staff members.
- This inconsistency results in poor data integrity, which directly hinders accurate clinical decision-making and automated risk assessment.

SOLUTION

AI-DRIVEN PREDICTIVE HEALTH MONITORING

- Continuously analyzes patient vital sign patterns to predict clinical deterioration before a crisis occurs

INSTANT EMERGENCY ALERT SYSTEM

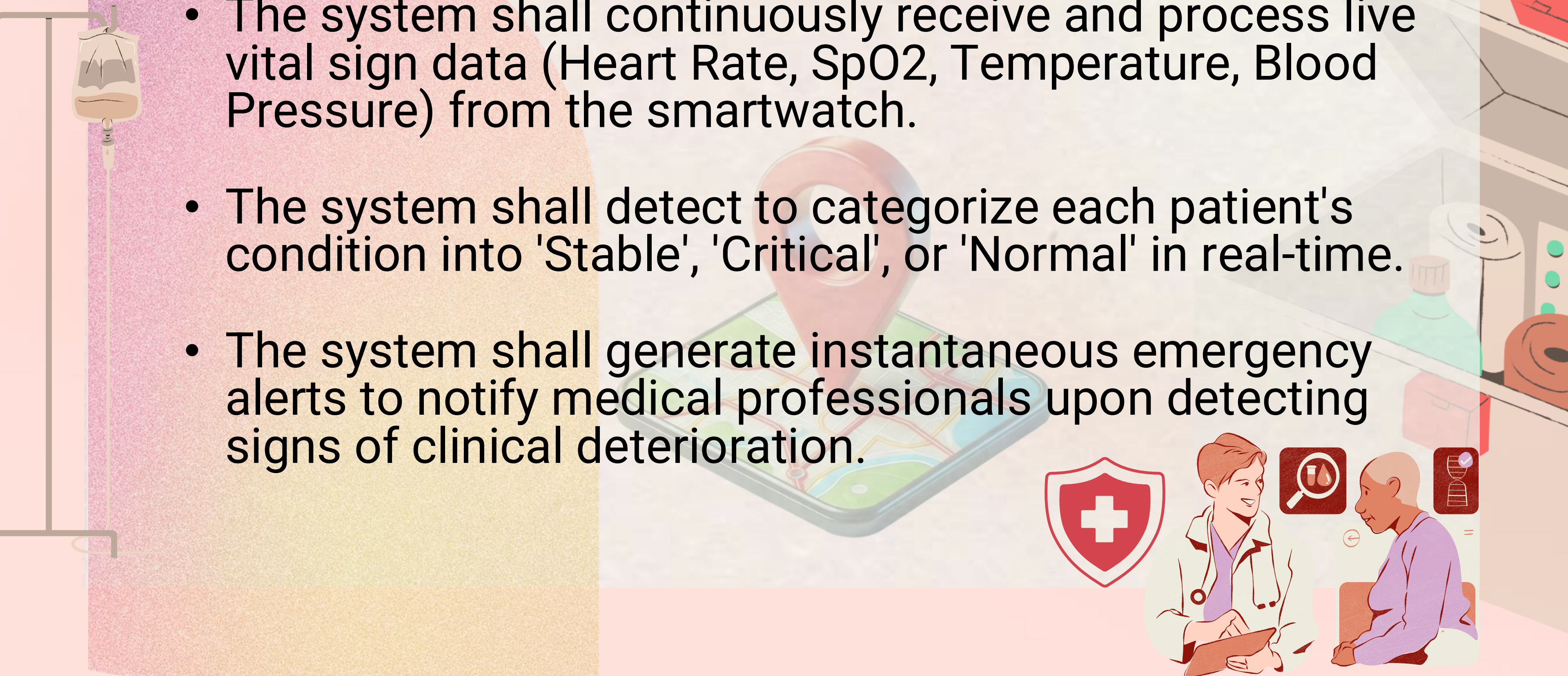
- Triggers immediate high-priority notifications to medical staff through mobile and web applications upon detecting anomalies.

CENTRALIZED HEALTH ANALYTICS DASHBOARD

- Visualizes real-time health status and historical trends for all patients on a single unified interface.

USER REQUIREMENTS

- The system shall continuously receive and process live vital sign data (Heart Rate, SpO2, Temperature, Blood Pressure) from the smartwatch.
- The system shall detect to categorize each patient's condition into 'Stable', 'Critical', or 'Normal' in real-time.
- The system shall generate instantaneous emergency alerts to notify medical professionals upon detecting signs of clinical deterioration.



DESIGN EXCELLENCE & CONTRIBUTION

This ML-driven engine continuously monitors patient vital signs and uses a Random Forest model to detect abnormal health conditions in real time. It automatically classifies patient status as Stable, Warning, or Emergency and generates instant alerts for critical situations.

```
--- Safe Band AI Multi-Level Risk Testing ---
```

```
>>> Analyzing Patient Sample 1:  
Predicted Status   : High Risk 🚨  
AI Confidence Level : 70.00%  
Probability Distribution: [Low: 0.03, Mid: 0.27, High: 0.70]
```

```
>>> Analyzing Patient Sample 2:  
Predicted Status   : Mid Risk ⚠️  
AI Confidence Level : 73.00%  
Probability Distribution: [Low: 0.12, Mid: 0.73, High: 0.15]
```

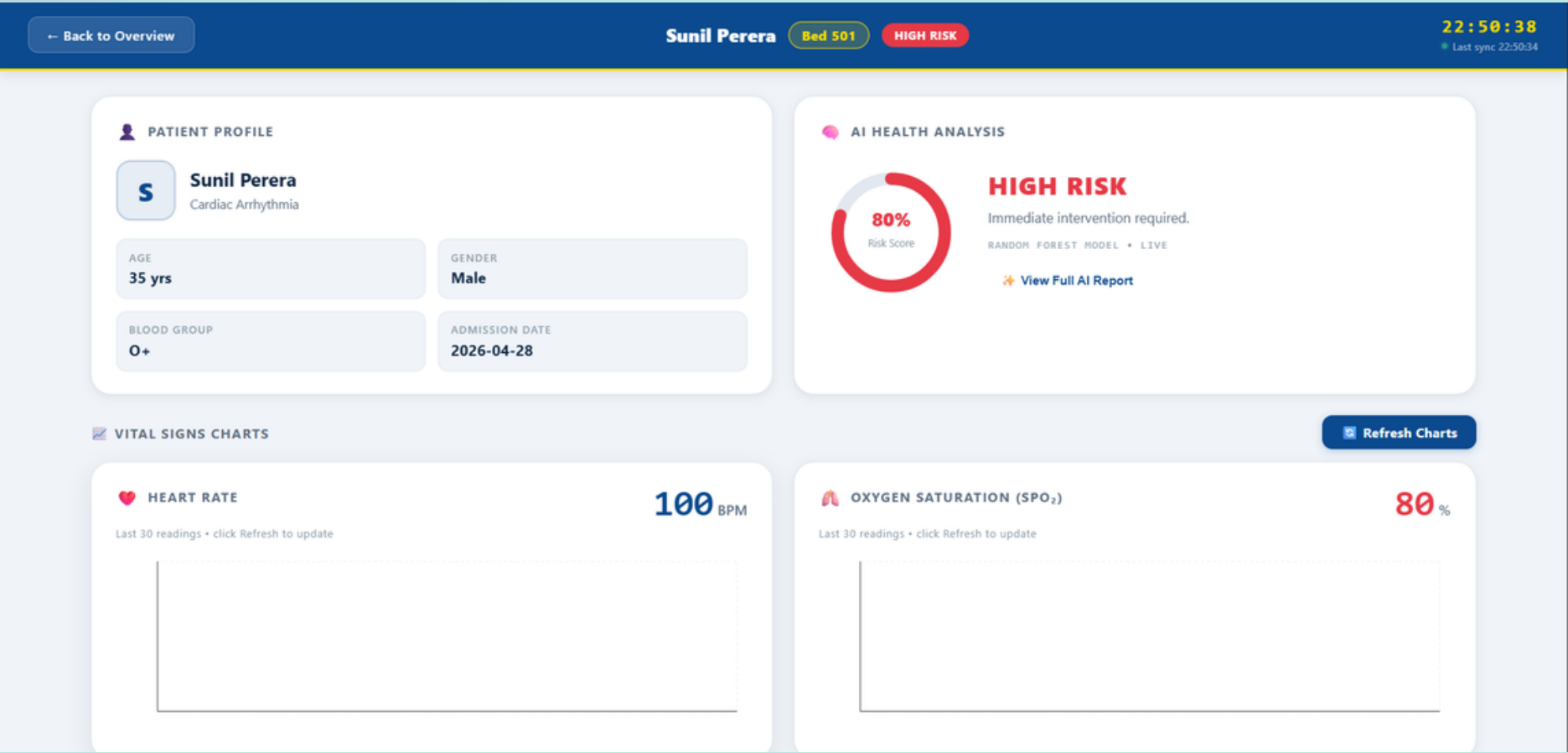
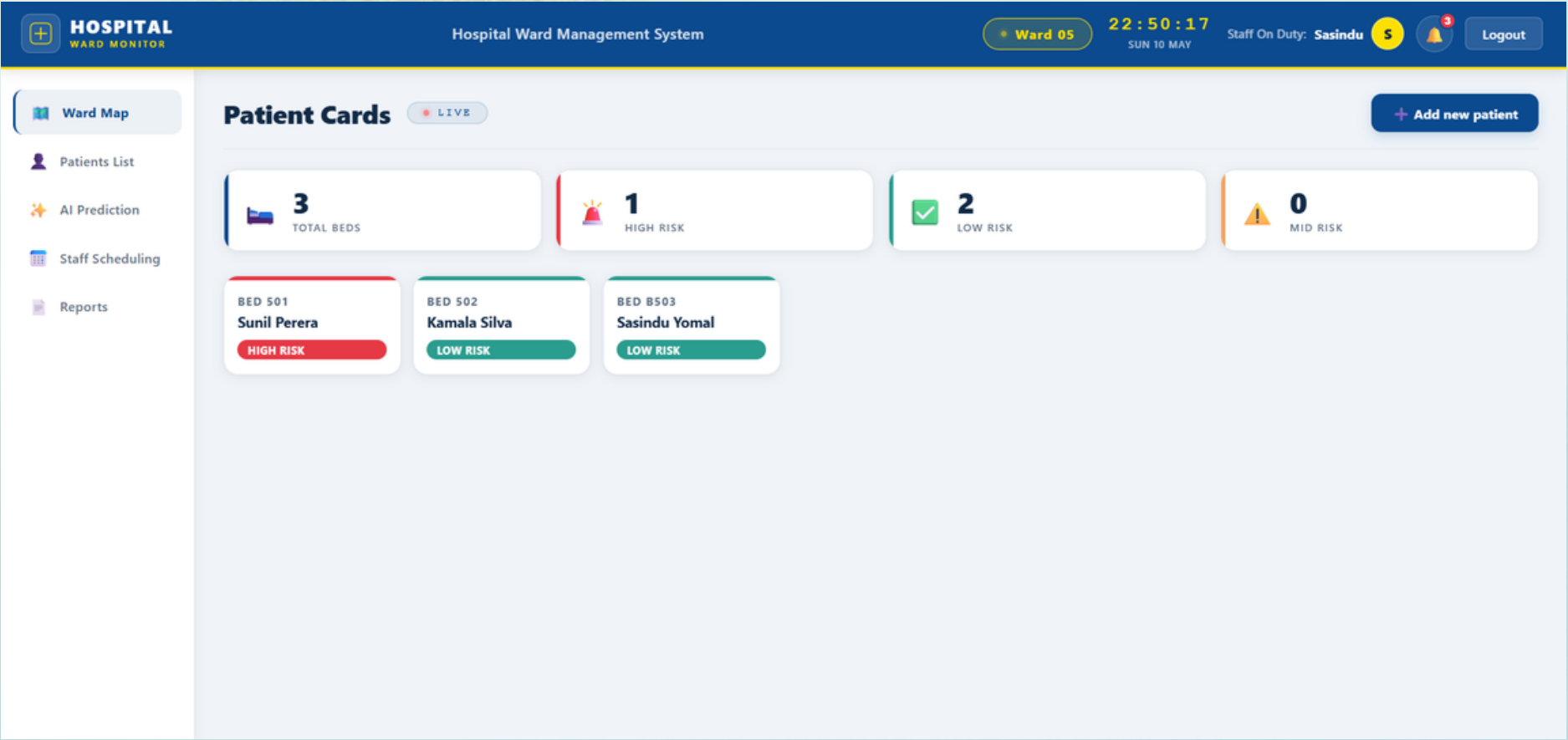
```
>>> Analyzing Patient Sample 3:  
Predicted Status   : Low Risk ✅  
AI Confidence Level : 97.00%  
Probability Distribution: [Low: 0.97, Mid: 0.02, High: 0.01]
```

```
=====  
--- Overall Model Performance Evaluation ---  
Overall Test Accuracy: 100.00%
```

```
--- Detailed Classification Report ---
```

	precision	recall	f1-score	support
Low Risk	1.00	1.00	1.00	1
Mid Risk	1.00	1.00	1.00	1
High Risk	1.00	1.00	1.00	1
accuracy			1.00	3
macro avg	1.00	1.00	1.00	3
weighted avg	1.00	1.00	1.00	3

DESIGN EXCELLENCE & CONTRIBUTION

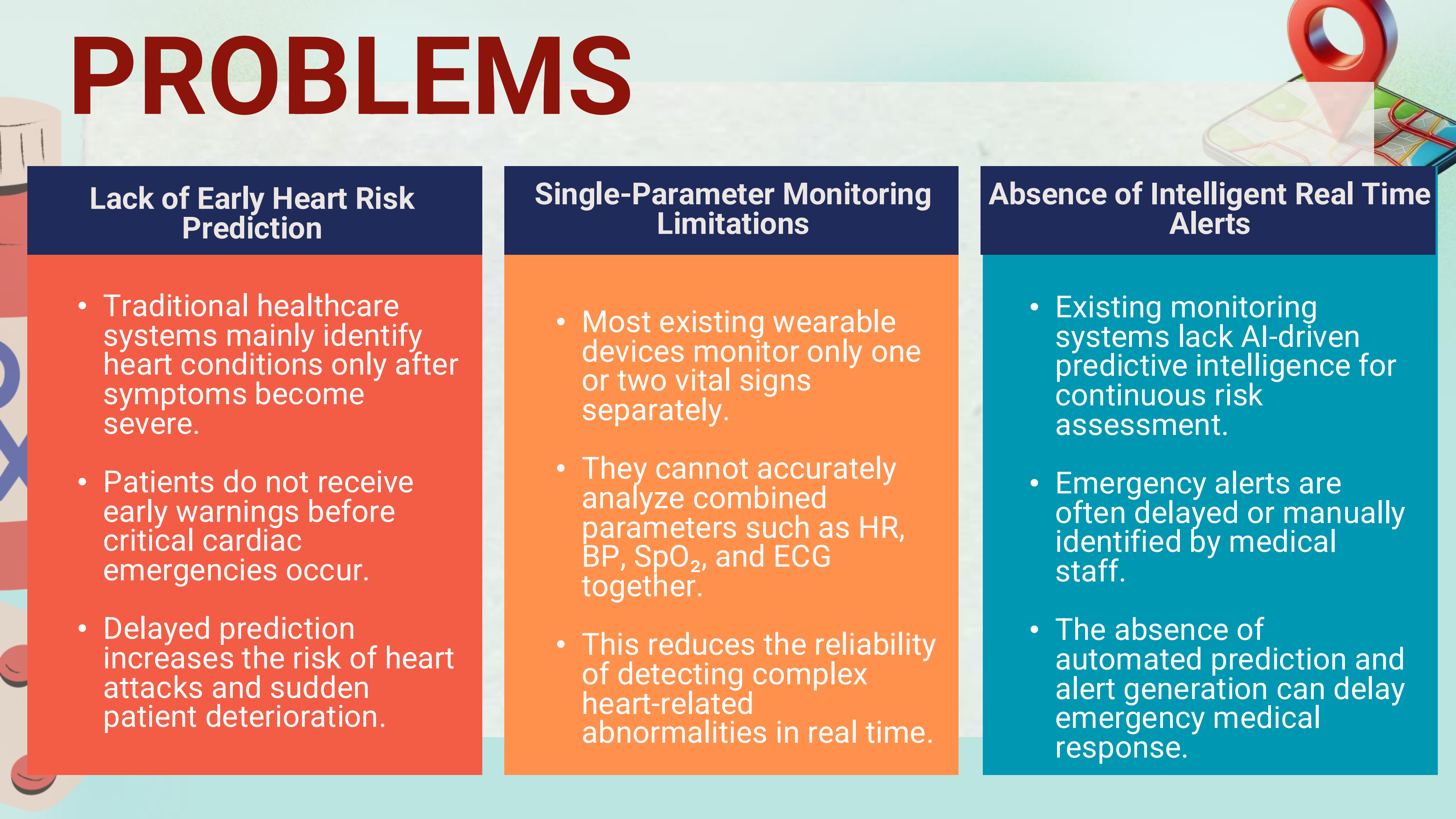


HEART RISK PREDICTION

IT22641038 | Gunawardana G.H.N | R26-IT-057



PROBLEMS



Lack of Early Heart Risk Prediction

- Traditional healthcare systems mainly identify heart conditions only after symptoms become severe.
- Patients do not receive early warnings before critical cardiac emergencies occur.
- Delayed prediction increases the risk of heart attacks and sudden patient deterioration.

Single-Parameter Monitoring Limitations

- Most existing wearable devices monitor only one or two vital signs separately.
- They cannot accurately analyze combined parameters such as HR, BP, SpO₂, and ECG together.
- This reduces the reliability of detecting complex heart-related abnormalities in real time.

Absence of Intelligent Real Time Alerts

- Existing monitoring systems lack AI-driven predictive intelligence for continuous risk assessment.
- Emergency alerts are often delayed or manually identified by medical staff.
- The absence of automated prediction and alert generation can delay emergency medical response.

SOLUTION

AI-BASED HEART RISK PREDICTION

- Predicts heart risks before emergencies occur
- Detects abnormal patient conditions using ML analysis

PATIENT RISK ALERT SYSTEM

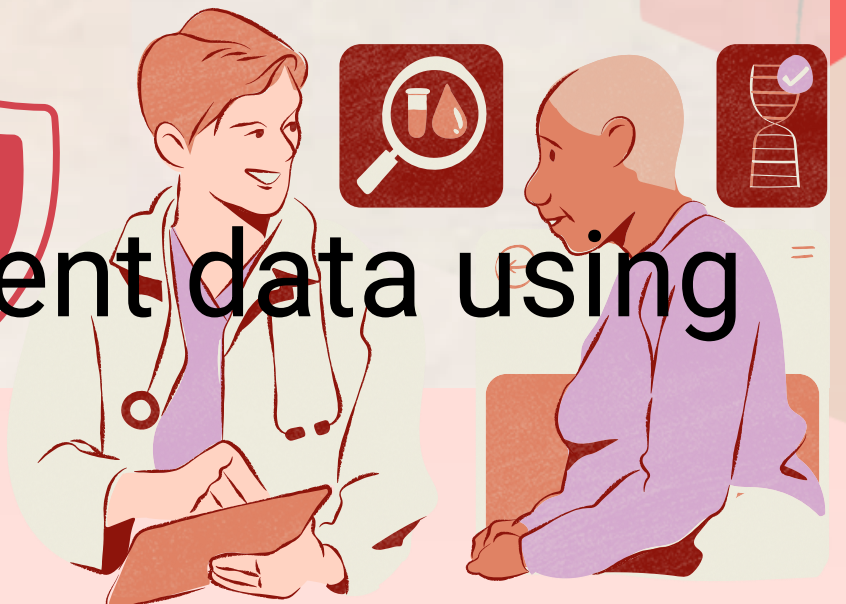
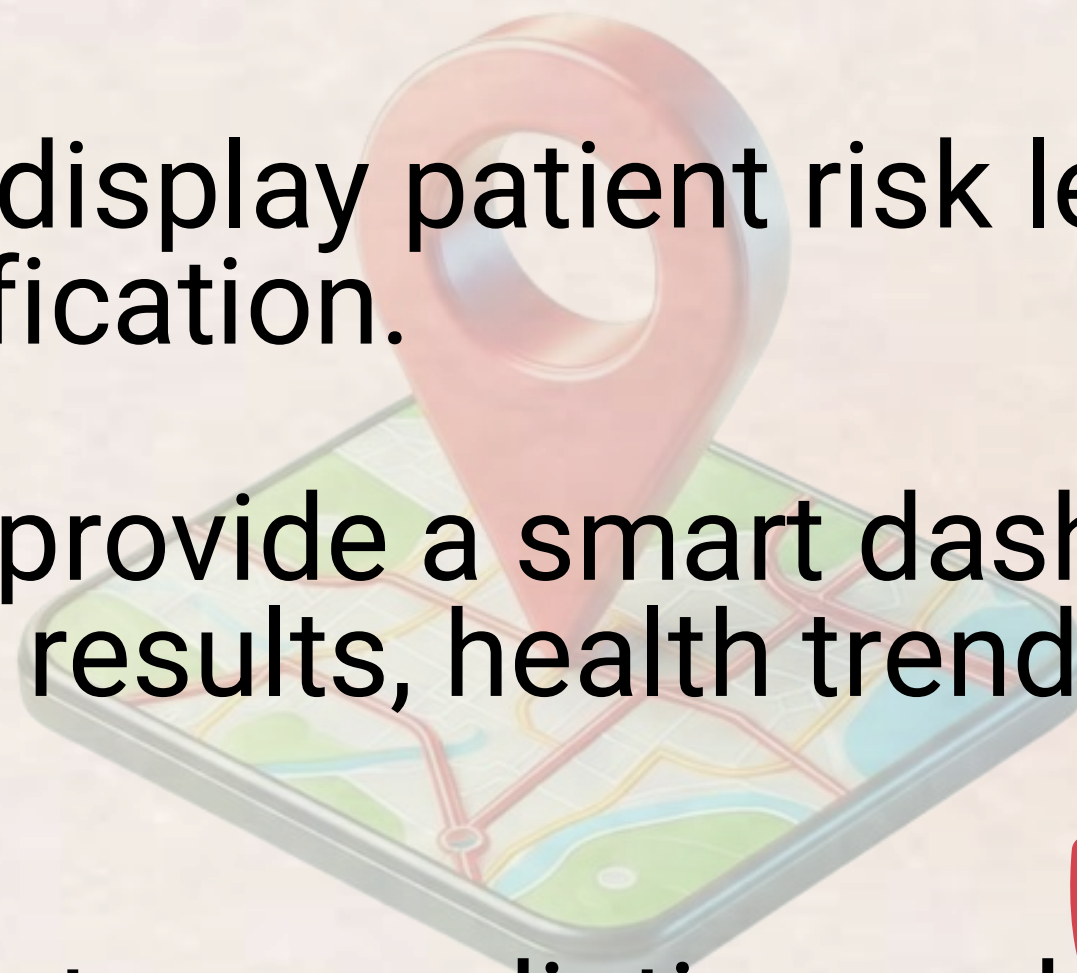
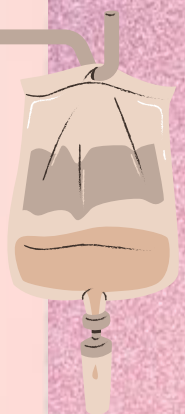
- Displays high-risk patient conditions
- Helps reduce emergency response delays

SMART HEALTH DASHBOARD

- Shows prediction results and trends
- Centralized monitoring system for healthcare staff

USER REQUIREMENTS

- The system shall predict patient heart risk levels as “Low Risk”, “Medium Risk”, or “High Risk” based on analyzed health data.
- The system shall display patient risk levels to support emergency identification.
- The system shall provide a smart dashboard to display patient prediction results, health trends, and risk information.
- The system shall store prediction and patient data using Firebase for monitoring purposes.



DESIGN EXCELLENCE & CONTRIBUTION

This ML-based heart risk prediction model analyzes patient health datasets using a Decision Tree algorithm to identify potential heart risk conditions. The model processes multiple health factors and predicts whether a patient is at low or high risk based on the analyzed data.

```
[1]: import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
[2]: df = pd.read_csv("heart_attack_sample_data.csv")
df.head()
```

	Age	Gender	Height	Weight	Smoking	Exercise_Frequency	Diabetes	Hypertension	Cholesterol_Level	Hereditary	Heart_Attack_Risk
0	58	Female	162	81	Yes	Daily	Yes	Yes	317	No	1
1	71	Male	182	62	Yes	Daily	Yes	Yes	233	Yes	1
2	48	Male	183	53	Yes	Rarely	No	Yes	262	Yes	1
3	34	Female	184	97	No	Rarely	Yes	Yes	225	No	0
4	62	Male	150	88	Yes	Never	No	Yes	271	No	1

```
[3]: df.info()
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 11 columns):
 #   Column              Non-Null Count  Dtype  
---  --
 0   Age                  500 non-null   int64  
 1   Gender                500 non-null   str     
 2   Height                500 non-null   int64  
 3   Weight                500 non-null   int64  
 4   Smoking               500 non-null   str     
 5   Exercise_Frequency    500 non-null   str     
 6   Diabetes              500 non-null   str     
 7   Hypertension          500 non-null   str     
 8   Cholesterol_Level     500 non-null   int64  
 9   Hereditary            500 non-null   str     
10   Heart_Attack_Risk     500 non-null   int64  
dtypes: int64(5), str(6)
memory usage: 43.1 KB
```

	Age	Height	Weight	Cholesterol_Level	Heart_Attack_Risk
count	500.000000	500.000000	500.000000	500.000000	500.000000
mean	50.396000	169.414000	81.750000	235.780000	0.860000
std	17.335982	11.932786	21.297386	50.111015	0.347335
min	20.000000	150.000000	45.000000	150.000000	0.000000
25%	36.750000	159.000000	62.750000	192.500000	1.000000

```
max    79.000000    189.000000    119.000000    319.000000    1.000000
```

```
[4]: df["Gender"] = df["Gender"].map({"Male":1, "Female":0})
df["Smoking"] = df["Smoking"].map({"Yes":1, "No":0})
df["Diabetes"] = df["Diabetes"].map({"Yes":1, "No":0})
df["Hypertension"] = df["Hypertension"].map({"Yes":1, "No":0})
df["Hereditary"] = df["Hereditary"].map({"Yes":1, "No":0})
df["Cholesterol_Level"] = df["Cholesterol_Level"].map({
    "Normal":0,
    "Borderline":1,
    "High":2
})
df["Exercise_Frequency"] = df["Exercise_Frequency"].map({
    "Rarely":0,
    "Monthly":1,
    "Weekly":2,
    "Daily":3
})
```

```
[5]: X = df.drop("Heart_Attack_Risk", axis=1)
y = df["Heart_Attack_Risk"]
```

```
[6]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
[7]: model = DecisionTreeClassifier(max_depth=5, random_state=42)
model.fit(X_train, y_train)
```

```
[7]: DecisionTreeClassifier
Parameters
```

```
[8]: pred = model.predict(X_test)
```

```
[9]: acc = accuracy_score(y_test, pred)
print("Accuracy:", acc)

Accuracy: 0.85
```

```
[10]: print(confusion_matrix(y_test, pred))
print(classification_report(y_test, pred))

[[10  8]
 [ 7 75]]
```

```
precision    recall    f1-score   support
```

```
[10]: print(confusion_matrix(y_test, pred))
print(classification_report(y_test, pred))

[[10  8]
 [ 7 75]]
```

		precision	recall	f1-score	support
	0	0.59	0.56	0.57	18
	1	0.90	0.91	0.91	82
accuracy				0.85	100
macro avg		0.75	0.74	0.74	100
weighted avg		0.85	0.85	0.85	100

```
[11]: import joblib

joblib.dump(model, "trained_heart_attack_model.pkl")
```

```
[11]: ['trained_heart_attack_model.pkl']
```

```
[12]: loaded_model = joblib.load("trained_heart_attack_model.pkl")
```

```
[13]: sample_data = [[
    55, # Age
    1, # Gender (Male=1 Female=0)
    170, # Height
    82, # Weight
    1, # Smoking (Yes=1)
    1, # Exercise_Frequency
    1, # Diabetes
    1, # Hypertension
    2, # Cholesterol_Level (High=2)
    1 # Hereditary
]]
```

```
[14]: prediction = loaded_model.predict(sample_data)
print(prediction)
```

```
[1]

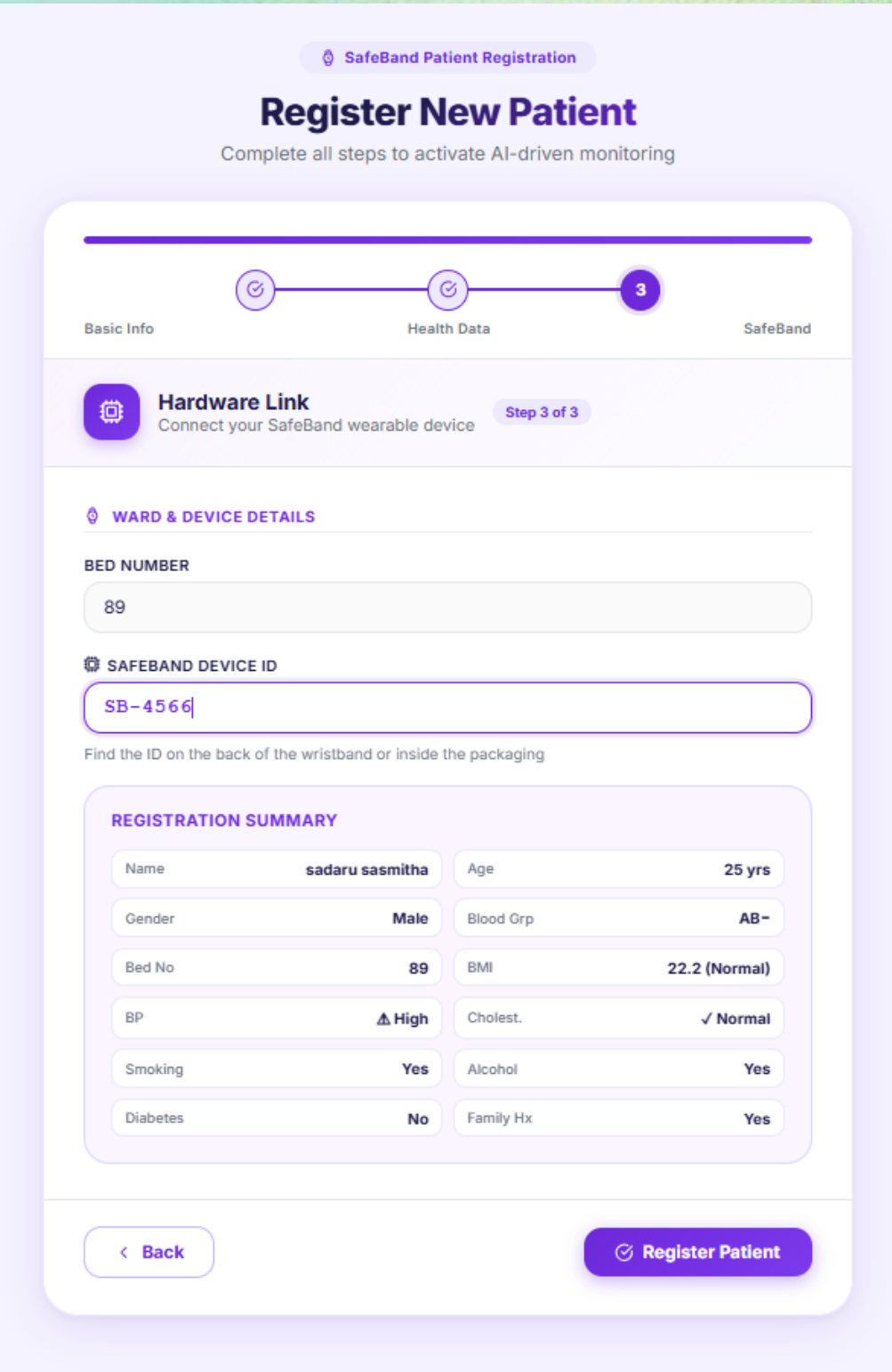
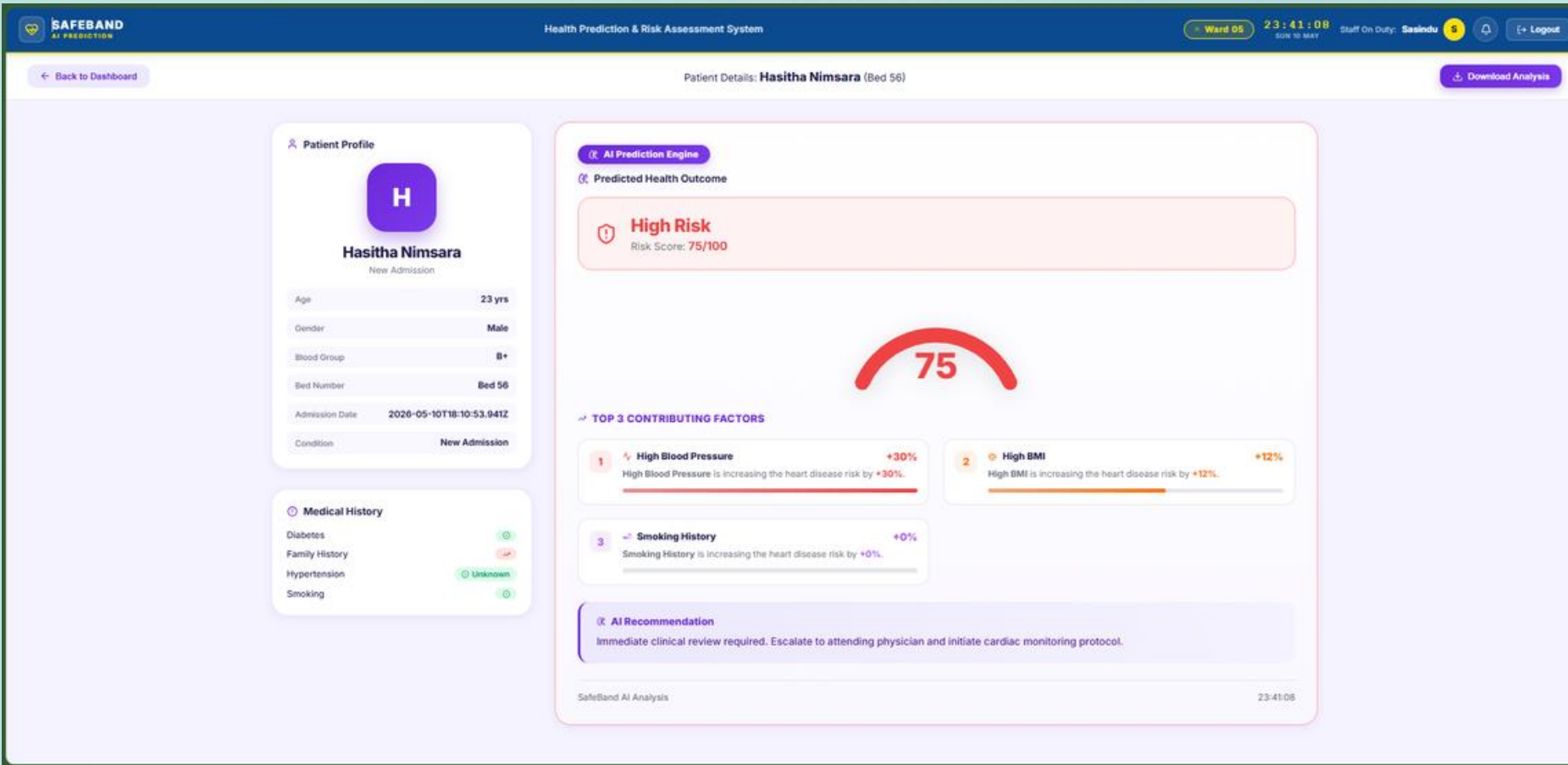
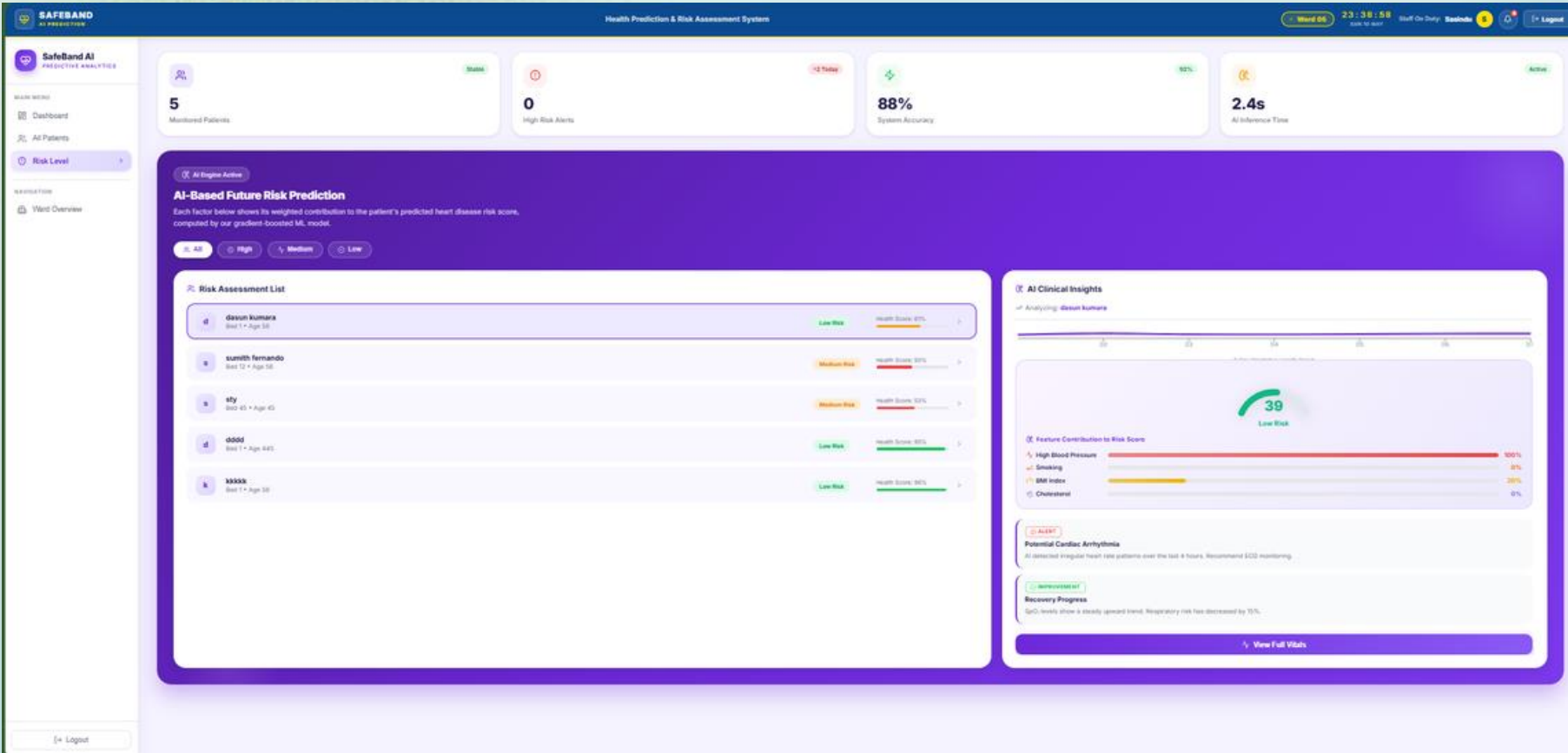
C:\Users\futur\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\sklearn\utils\validation.
names, but DecisionTreeClassifier was fitted with feature names
warnings.warn(
```

```
[15]: if prediction[0] == 1:
    print("Patient is HIGH RISK")
else:
    print("Patient is LOW RISK")
```

```
Patient is HIGH RISK
```

```
[ ]:
```

DESIGN EXCELLENCE & CONTRIBUTION





THANK YOU

R26-IT-057